UNIT 5 · TOPIC 1

Anthropometry in Equipment Design

Syllabus topic: Anthropometry in equipment design

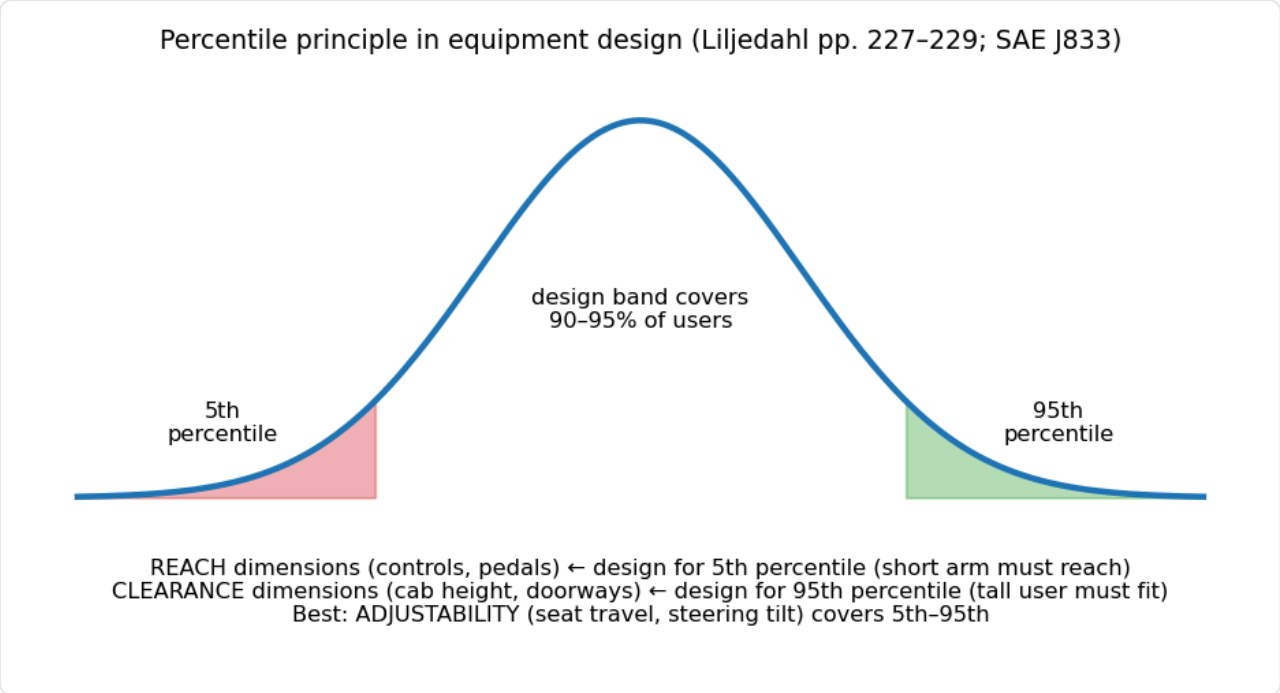
Ergonomics and human factors — scope

Ergonomics (human factors) fits the machine and the task to the human, so the operator can carry out many complex tasks efficiently and safely with minimum fatigue. The factors include riding comfort, visibility, location/arrangement and ease of controls, thermal comfort and sound control (Liljedahl p. 203).

Two data pillars:

- **ANTHROPOMETRY** — body dimensions and strengths (this topic).
- **WORK PHYSIOLOGY** — energy cost and body responses (Topic 2).

The percentile principle (Liljedahl pp. 227–230)



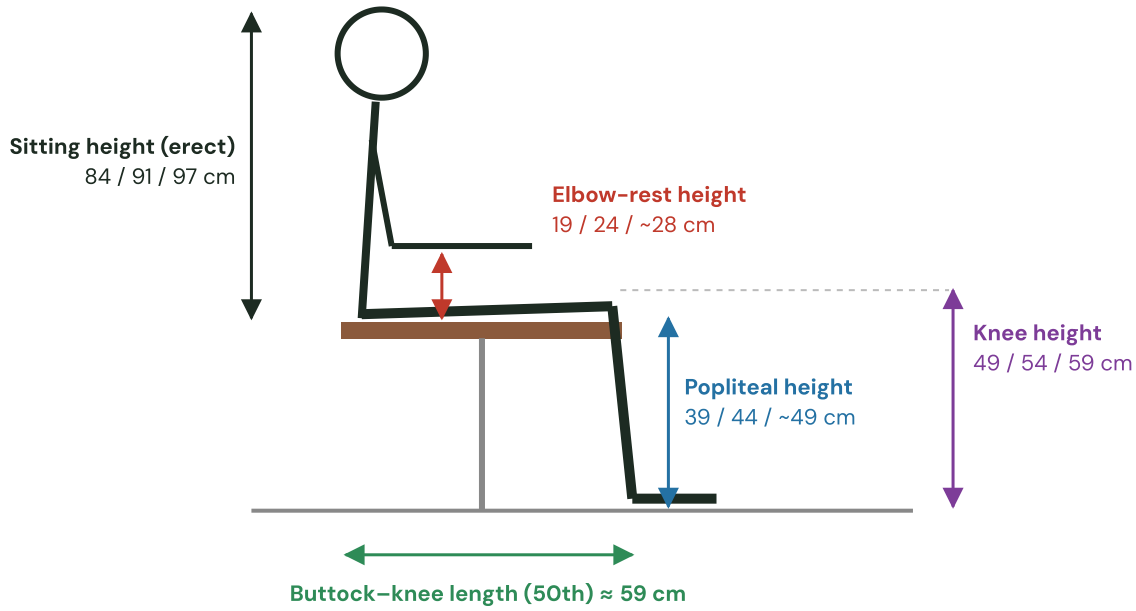
Percentile principle in equipment design

Anthropometric data are tabulated as 5th / 50th / 95th percentiles; workplace design normally uses the 95th and 5th percentile values (p. 229).

Dimension type	Design percentile	Example
CLEARANCE (cab height, hatches, doorways)	UPPER — 95th	Inside cab height 185 cm keeps 95% of US male drivers from bumping their heads (p. 227); women's 95th-percentile standing height ≈ 170 cm
REACH (controls a short arm must grasp)	LOWER — 5th	Pedals and levers (p. 229)
Range 2.5–97.5 percentile	—	Includes 95% of the population (p. 229)
ADJUSTABILITY (seat fore-aft/height travel, steering tilt)	Spans 5th–95th	Best practice

Key dimensions (SAE J833; Table 9-5, p. 228)

Selected SAE J833 male data, 5th / 50th / 95th percentile (Liljedahl Table 9-5); 50th-percentile mass ≈ 75 kg



Key seated anthropometric dimensions for workplace design (redrawn from the source figure)

Male dimension (cm)	5th	50th	95th
Stature	162	173	185
Sitting height, erect	84	91	97
Knee height	49	54	59
Popliteal height	39	44	~49
Elbow-rest height	19	24	~28
Buttock–knee length	—	59	—
Seat breadth	—	36	—
Mass (kg)	—	75	—

Body dimension	Sets the design of
Popliteal height	Seat height
Buttock–popliteal length	Seat-pan depth
Elbow-rest height	Armrest / control datum
Elbow-to-elbow breadth	Lateral spacing

Indian context

- Indian operators are smaller than the US SAE population.
- The CIAE/AICRP ergonomics programme compiled an **Indian agricultural-worker anthropometric database** — men AND women, dozens of body dimensions plus strength parameters (grip, push-pull, leg strength). It is the design basis for Indian hand tools and machines; western data oversize controls and heights.
- **Strength data set force limits:** sustained hand-crank/pedal loads stay well below maximum voluntary strength — design ≤ ~**30%** for continuous work.

Practice MCQs

Q1. Anthropometry is the science of:

(a) Muscle chemistry
(b) Human body dimensions and strengths
(c) Soil physics
(d) Plant growth

Q2. Standard workplace-design practice uses which percentiles?

(a) 25th & 75th
(b) 5th & 95th
(c) 50th only
(d) 1st & 99th

Q3. Clearance dimensions (cab height, doorways) are designed for the:

(a) 5th percentile
(b) 50th
(c) 95th percentile
(d) Mean

Q4. Reach dimensions (controls, pedals) are designed for the:

(a) 5th percentile
(b) 95th
(c) 99th
(d) Tallest user

Q5. Per Liljedahl (p. 227), inside cab height to protect 95% of male drivers is:

(a) 160 cm
(b) 170 cm
(c) 185 cm
(d) 200 cm

Q6. The recommended anthropometric practice for agricultural equipment in the US is:

(a) SAE J833
(b) IS 4468
(c) ISO 500
(d) OECD

Q7. Seat height in workplace design is based on:

(a) Stature
(b) Popliteal height
(c) Elbow breadth
(d) Mass

Q8. Seat pan depth is based on:

(a) Buttock–popliteal length
(b) Knee height
(c) Sitting height
(d) Stature

Q9. The 50th-percentile mass of the SAE male population (Table 9-5) is about:

(a) 55 kg (b) 65 kg
(c) 75 kg (d) 90 kg

Q10. The best design strategy for spanning users' sizes is:

(a) Design for the average (b) Adjustability from 5th to 95th percentile
(c) One size (d) Design for tallest

Q11. Indian equipment design uses CIAE/AICRP anthropometric data because:

(a) US data are secret (b) Indian body dimensions are smaller — western data oversize designs
(c) Fashion (d) Cost

Q12. For continuous work, sustained force should not exceed about what share of maximum voluntary strength?

(a) 90% (b) 60%
(c) 30% (d) 5%

Answers

Q1 — (b) Human body dimensions and strengths
Q2 — (b) 5th & 95th
Q3 — (c) 95th percentile — *Liljedahl pp. 227–229 — clearance from the upper percentile.*
Q4 — (a) 5th percentile — *Liljedahl p. 229 — reach from the lower percentile.*
Q5 — (c) 185 cm — *Liljedahl p. 227 — the 185-cm cab-height example.*
Q6 — (a) SAE J833
Q7 — (b) Popliteal height — *Table 9-5, p. 228 (SAE J833 data).*
Q8 — (a) Buttock–popliteal length
Q9 — (c) 75 kg — *Table 9-5, p. 228 — 50th-percentile mass 75 kg.*
Q10 — (b) Adjustability from 5th to 95th percentile
Q11 — (b) Indian body dimensions are smaller — western data oversize designs
Q12 — (c) 30%

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 "Human Factors in Tractor Design", pp. 203–246; CIAE Bhopal / AICRP-Ergonomics literature (Gite & Yadav) for Indian physiological norms; SAE J833; ASHRAE comfort work; NPTEL ergonomics lectures.